Lab 5 Postlab: Metal Evaporation (Flow Cytometer)

Run details. Al was thermally evaporated from a tungsten boat at about 10^{-6} torr in a bell jar about 450 mm across and 600 mm tall. We used SPR220 photoresist instead of the SJR5740 listed in the handout, spun at 2000 rpm for 60 s and soft-baked at about 105 °C for 7 min. At the end of the run, the STC-200 deposition rate controller read THICK = 0.464 kÅ.

Fill in before submitting: the Q1 estimate assumes we loaded **4 pieces of 1/2" Al wire, 1.0 mm in diameter**. If you loaded a different amount, take your value from the table in Q1.

1. Aluminum thickness: our estimate vs. the gauge

Gauge reading. The STC-200 showed 0.464 kÅ = 464 Å = **46.4 nm**.

Mass of Al loaded. Each piece is a 1/2" (1.27 cm) length of 1.0 mm-diameter wire:

$$m_{piece} = \rho \pi r^2 L = (2.70\ \text{g/cm}^3) \times \pi \times (0.05\ \text{cm})^2 \times (1.27\ \text{cm}) = 0.0269\ \text{g}$$

Four pieces give m ≈ 0.108 g. The procedure says to evaporate until the boat is empty, so we take all of it as evaporated.

Geometry. A flat boat acts as a small surface source. It emits with a cosine distribution, mostly upward. For a wafer directly above the source at distance h, the film thickness is:

$$t = m / (\pi \rho h^2)$$

The wafers sit near the top of the 600 mm bell jar, so take h ≈ 60 cm:

$$t = 0.108\ \text{g} / [\pi \times (2.70\ \text{g/cm}^3) \times (60\ \text{cm})^2] = 3.5 \times 10^{-6}\ \text{cm} \approx \mathbf{35\ nm\ (350\ Å)}$$

Pieces of 1/2" wire	1.0 mm wire: mass → thickness	1.5 mm wire: mass → thickness
2	0.054 g → 18 nm	0.121 g → 40 nm
3	0.081 g → 26 nm	0.182 g → 60 nm
4	**0.108 g → 35 nm**	0.242 g → 79 nm
5	0.135 g → 44 nm	0.303 g → 99 nm
6	0.162 g → 53 nm	0.364 g → 119 nm

Check against the chamber size. Suppose instead that the Al coats the whole inside of the bell jar evenly. With diameter 45 cm and height 60 cm, the area is $\pi r^2 + 2\pi rh \approx 1.0 \times 10^4\ \text{cm}^2$, and 0.108 g gives about **40 nm**. That is close to the estimate above. (If the 450 mm were the radius rather than the diameter, the area would double and the film would be about 17 nm.)

Uniformity. With this geometry, the film at a point x away from the wafer centre is $t(x)/t_0 = [1 + (x/h)^2]^{-2}$. At the edge of a 100 mm wafer (x = 5 cm) this is 0.986, so the Al thickness varies by only about 1–2% across the wafer.

Comparison. Our estimate (about 35 nm) and the gauge (46.4 nm) agree to within about 25–30%. For the gauge to be exactly right, we would have needed $m = \pi\rho h^2 t \approx 0.14$ g of Al, roughly 5 pieces of 1 mm wire. Reasons the two numbers differ:

- **The amount loaded is uncertain.** Wire diameter and piece length were not measured. Part of the Al also wets or alloys with the hot tungsten boat and never evaporates.
- **Simplified geometry.** The boat is neither a perfect point source nor a perfect surface source. A pure point source ($t = m / 4\pi\rho h^2$) would give only about 9 nm, so the real emission pattern matters a lot. The source-to-wafer distance was also estimated from the chamber height.
- **The gauge's own settings.** The quartz crystal monitor works out thickness from mass, using the density (2.70 g/cm^3 for Al), the Z-ratio (1.08 for Al), and a tooling factor. The tooling factor corrects for the crystal sitting closer to or farther from the source than the wafers. If the controller still had another film's settings, or the default tooling factor of 100%, its reading would be off by the same factor.
- **Film density.** Evaporated films can be slightly less dense than bulk Al, which makes them a little thicker than the bulk density predicts.

To measure the true thickness, use a profilometer step height after next week's Al etch.

2. Photoresist used and its thickness

Resist. We used **MEGAPOSIT SPR220** (DuPont, formerly Shipley / Rohm and Haas; sold in the US through Kayaku Advanced Materials) instead of the SJR5740 named in the handout. SPR220 is a thick, positive i-line resist made for 1–10 µm films. The number after the dash gives the nominal thickness: SPR220-3.0 is about 3 µm and SPR220-7.0 is about 7 µm.

Thickness. For spin coating, thickness falls as one over the square root of spin speed ($t \propto 1/\sqrt{\omega}$). SPR220-3.0 gives about 3.0 µm at 2500 rpm, so at 2000 rpm:

$$t \approx 3.0\ \mu\text{m} \times \sqrt{(2500/2000)} \approx 3.3\ \mu\text{m}$$

Cleanroom data report about 3.2 µm at 2000 rpm for SPR220-3.0. We therefore expect **about 3.2–3.4 µm** in the flat centre of the wafer. Soft-baking drives out solvent and shrinks the film by a few percent. (If the bottle was SPR220-7.0, the film would be about 7–8 µm at 2000 rpm. Check the label.)

3. Positive vs. negative photoresist

	Positive resist	Negative resist
Effect of exposure	The exposed areas become **more** soluble in developer (a photoactive compound breaks down, e.g. DNQ to a carboxylic acid in a novolac resin)	The exposed areas become **less** soluble (the polymer crosslinks or hardens)
What stays after developing	The **unexposed** areas, so the pattern is the **same** as the mask's opaque regions	The **exposed** areas, so the pattern is the **reverse** of the mask
Resolution	Better. The film does not swell in developer, so edges stay sharp	Older types swell in solvent developer, which limits resolution. Modern chemically amplified types are much better
Sidewalls and uses	Steep sidewalls. Good for etch masks. The standard choice for IC patterning	Can give undercut (re-entrant) profiles, which are good for lift-off. Can be very thick and tough (SU-8)
Chemical resistance and adhesion	Moderate	Usually stronger. Crosslinked films can be hard to strip

Resists used in industry today, and who makes them:

- **Positive:**
  - **DuPont (formerly Shipley / Dow):** MEGAPOSIT SPR220 (the one we used), S1813 / S1800 series, SPR955.
  - **Merck (EMD Performance Materials, AZ brand):** AZ 1512, AZ 4620, AZ 10XT.
  - Others: TOK (Tokyo Ohka Kogyo) THMR-iP series, and JSR.
- **Negative:**
  - **Kayaku Advanced Materials (formerly MicroChem):** SU-8 and KMPR.
  - **Merck (AZ brand):** AZ nLOF 2020 / 2070, a lift-off resist.
  - Others: Futurrex NR9 series, and micro resist technology ma-N 1400.

4. Is the photoresist the same thickness everywhere?

No. The resist is close to uniform across the flat centre of the wafer, but it varies in two places:

1. **Edge bead (thickest overall).** Resist piles up in a ridge within about 1–3 mm of the wafer rim, and near the flat. Two things cause it. Surface tension holds liquid at the edge as it is flung outward. Solvent also evaporates faster at the rim, so the resist there thickens before it can level out. The bead can be 2–5 times as thick as the film in the centre. That matters next week: it can stop the mask from sitting flat on the wafer during contact exposure.
2. **Over the device's surface relief.** Spun resist tends to level itself out. It is **thicker in low areas** (windows or trenches from earlier steps) and **thinner over raised features**, and thinnest right at the top corners of steps. Exposure and development times must be long enough to clear the thickest parts.

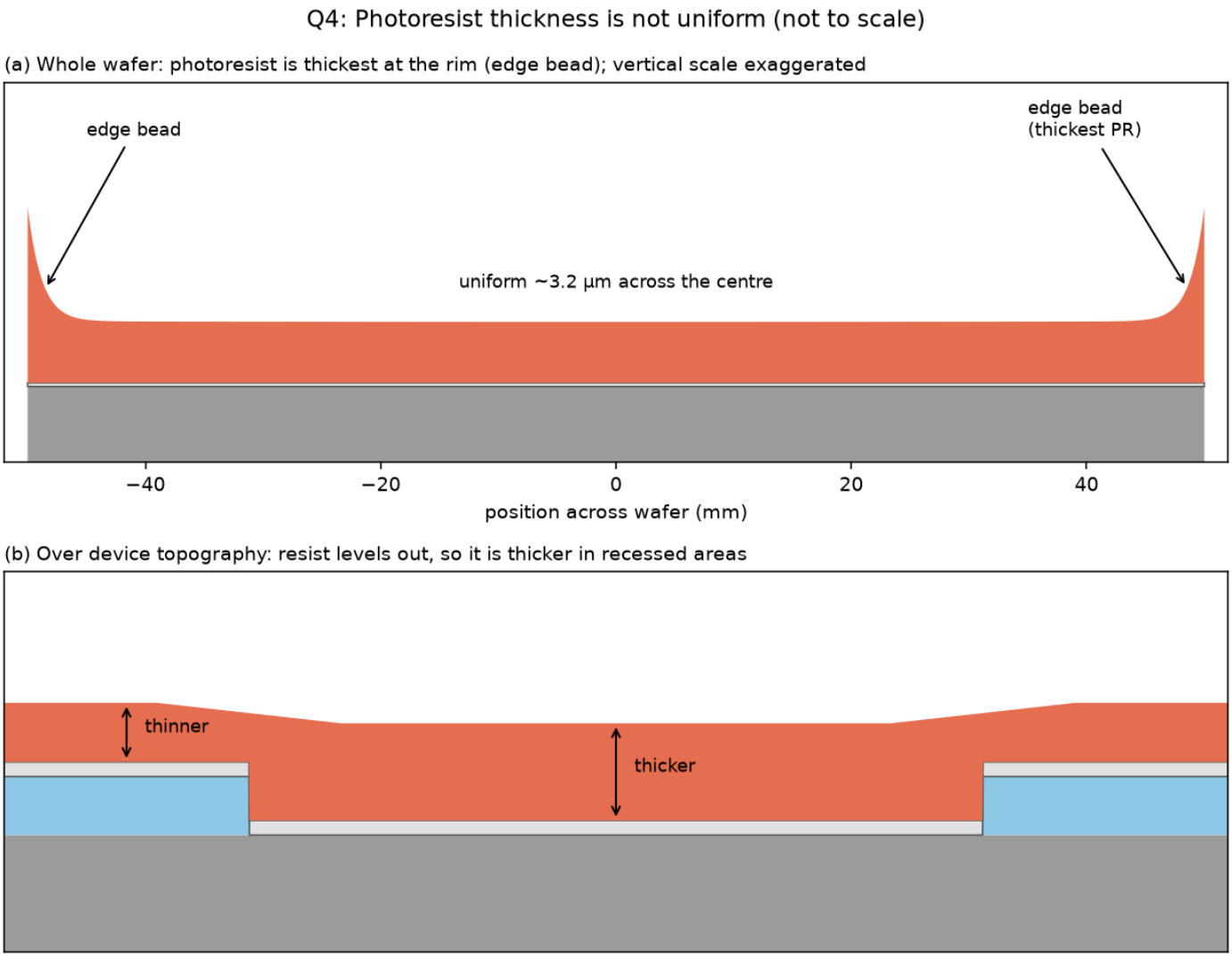


Figure 1. Photoresist thickness across the wafer (edge bead) and over device features. Not to scale.

5. Important cross-sections after this lab

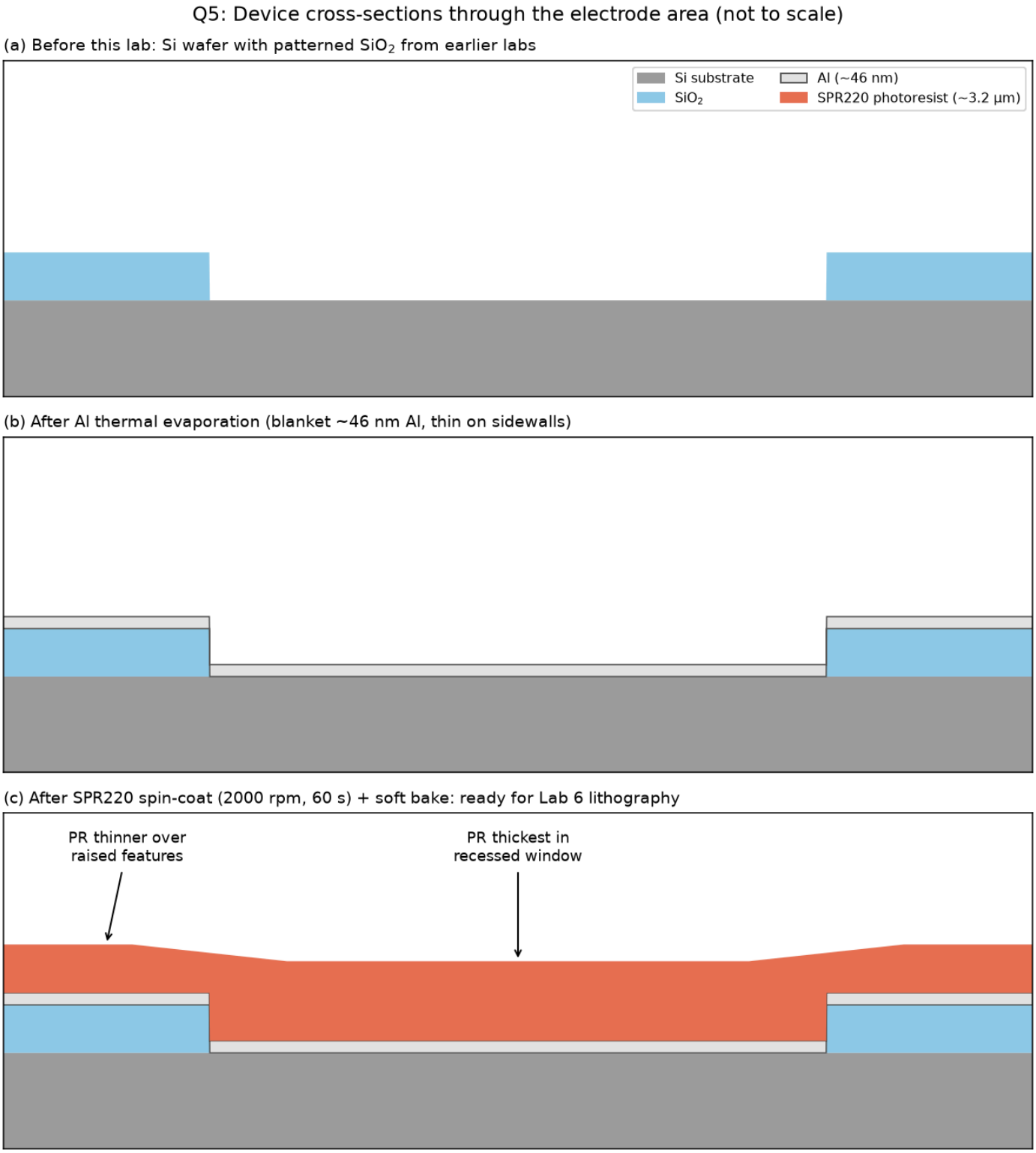


Figure 2. Device cross-sections through the electrode area. Not to scale.

- **(a) Starting point.** The Si wafer with the patterned SiO_2 from earlier labs.
- **(b) After evaporation.** A blanket Al film, about 46 nm thick, covers everything. Evaporation travels in straight lines, so the vertical sidewalls get very little Al. The Al may even break at tall steps.
- **(c) After spin-coating and soft bake.** About 3.2 µm of SPR220 covers the Al. It is thicker in recessed areas and builds up at the wafer edge. The wafer is now ready for next week's lithography, which will

pattern the Al into electrodes.

6. Why pump down to 1×10^{-6} torr before evaporating?

1. **Long mean free path, so atoms fly straight.** The mean free path of gas molecules is $\lambda = kT / (\sqrt{2}\,\pi d^2 P)$. At 10^{-6} torr (1.33×10^{-4} Pa) and room temperature, with $d \approx 0.37$ nm:

   $$\lambda = (1.38 \times 10^{-23} \times 295) / [\sqrt{2} \times \pi \times (3.7 \times 10^{-10})^2 \times 1.33 \times 10^{-4}] \approx 50 \text{ m}$$

   That is about 80 times the 0.6 m path from source to wafer. Only $1 - e^{-0.6/50} \approx 1\%$ of Al atoms hit a gas molecule on the way, so they arrive in a straight line. This gives a predictable, directional film. At higher pressure, the atoms would scatter, lose energy, and form a porous, poorly controlled deposit.
2. **Film purity and adhesion.** Even at 10^{-6} torr, background gas (mostly water vapour and O_2) can cover a surface with one molecular layer in about 1 s. At higher pressures this happens much faster. Aluminum reacts readily with oxygen and water, so a poor vacuum would leave oxide and hydroxide mixed into the film. That raises the resistivity of our electrodes and weakens adhesion. Lower pressure means fewer contaminants arriving compared with Al atoms.
3. **Protecting the hot source.** The tungsten boat and the molten Al run at over 1000 °C. With too much oxygen or water present, the boat and the charge would oxidize, which can burn out the boat and cause spitting.
4. **Stable process and equipment.** A clean, low base pressure makes the deposition rate repeatable. It is also needed for the ion gauge filament and the crystal monitor to work properly.

Sources

- University of Michigan LNF, CEE 200X spinner (SPR220 at 2000 rpm): https://lnf-wiki.eecs.umich.edu/wiki/index.php?title=CEE_200X_photoresist_spinner_1
- UCSB Nanofab, SPR-220-3.0 optimization: https://wiki.nanofab.ucsb.edu/w/images/0/01/SPR-220-3.0_OptimizationNew.pdf
- Kayaku Advanced Materials, MEGAPOSIT SPR220 series: https://kayakuam.com/products/megaposit-spr220-series-thick-i-line-photoresists/
- SPR220 data sheet (UPenn copy): https://www.seas.upenn.edu/~nanosop/documents/SPR_220_Data_Sheet.pdf